

CMRVision: A Foundation Model for Cardiac MR Image Analysis



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Objective: To train a CMR-specific foundation model and evaluate architectural and training design choices.

Motivation

- Cardiac magnetic resonance (CMR) is the reference standard for noninvasive assessment of cardiac anatomy, function, and tissue characterization.
- Typical CMR examination comprises multiple sequences (e.g., cine, late gadolinium enhancement, and parametric mapping) acquired across different anatomical views.
- Self-supervised foundation models (FM) have been increasingly explored for CMR and show strong performance; systematic investigations of FMs spanning multiple CMR sequences remain limited.

Methods

Pretraining

- Train a ViT-S backbone on unlabeled CMR data using a DINOv3-style [1] SSL framework combining four objectives:
 - (i) a DINO loss on global embeddings, enforcing student–teacher consistency across views;
 - (ii) an iBOT patch-level loss aligning local patch embeddings for fine-grained anatomical representation;
 - (iii) KoLeo regularization encouraging a uniform embedding distribution and preventing collapse;
 - (iv) Gram anchoring aligning second-order patch correlations to preserve CMR-specific inter-patch relationships and spatial coherence.
- Adaptations:
 - (i) a smaller patch size (8×8) for fine-grained structures;
 - (ii) early Gram anchoring, enabled by NI initialization;
 - (iii) two-stage training (half- then full-resolution) for efficiency;
 - (iv) domain-specific augmentations, removing operations unsuitable for CMR (e.g., color jitter, vertical flips); and
 - (v) an adjusted crop scale (minimum 0.1) for anatomically meaningful local views.

Downstream Tasks and Evaluation

- Unified multi-task, multi-sequence segmentation:** cine short-axis (SAX), cine long-axis (LAX), LGE SAX, and parametric mapping SAX
- Cine view classification:** 6 classes including commonly available (SAX, 2CH, 3CH, 4CH) and rarer classes (Aorta, Others)
- Zero-shot segmentation:** LGE LAX – an unseen view-sequence combination with lower SNR, heterogeneous inversion-recovery contrast, and irregular myocardial boundaries from scar enhancement

Data & Implementation

Pretraining

- 36 million images from 27,000 studies on multiple 1.5 T and 3 T scanners, spanning diverse sequences, views, and patient populations
- ViT-S encoder, patch size 8, image size 224 × 224, half-resolution training followed by full resolution (3 days each)

Downstream Tasks

- Segmentation:** 1,873 patients and 11,022 images
 - Cine SAX (ACDC [16]): LV, myocardium, RV; 150 patients / 300 images (balanced SSFP cine).
 - Cine LAX (Kaggle [18]): LV, myocardium, LA, RA; 436 patients / 874 images, 4-chamber (4CH) view only.
 - LGE SAX (EMIDEC [17] + Centers 1 & 2): LV, myocardium; 1,143 patients / 8,582 images (T1-weighted PSIR and inversion-recovery FLASH).
 - Mapping SAX (Center 3): LV, myocardium; 144 patients / 1,266 images (pre- and post-contrast T1 and T2 mapping).
 - UNETR [8] architecture, end-to-end finetuning

- Classification:** 2,036 images from 147 patients (1,511/218/307 images for train/val/test). View distribution: SAX (41.1%), “other” (23.1%), 2-chamber (11.2%), 4-chamber (9.3%), aorta (8.9%), 3-chamber (6.5%). Encoders were appended with a 1-layer linear classifier for end-to-end finetuning

- Zero shot segmentation:** Myocardium on Center 1; 22 patients / 57 images, 2-chamber (2CH) and 4CH views.

Conclusions

- CMRVision learns more transferable representations than natural-image-pretrained models, with consistent performance across segmentation, classification, and zero-shot transfer.
- Smaller patch sizes and patch-level objectives improve performance, highlighting the importance of domain-specific design for capturing fine-grained cardiac features.
- Limitations: Evaluation is restricted to CMR and DINO-family objectives, with limited exploration of fine-tuning strategies beyond LoRA. The benchmark covers 11,022 images, although some task-specific subsets remain relatively small.

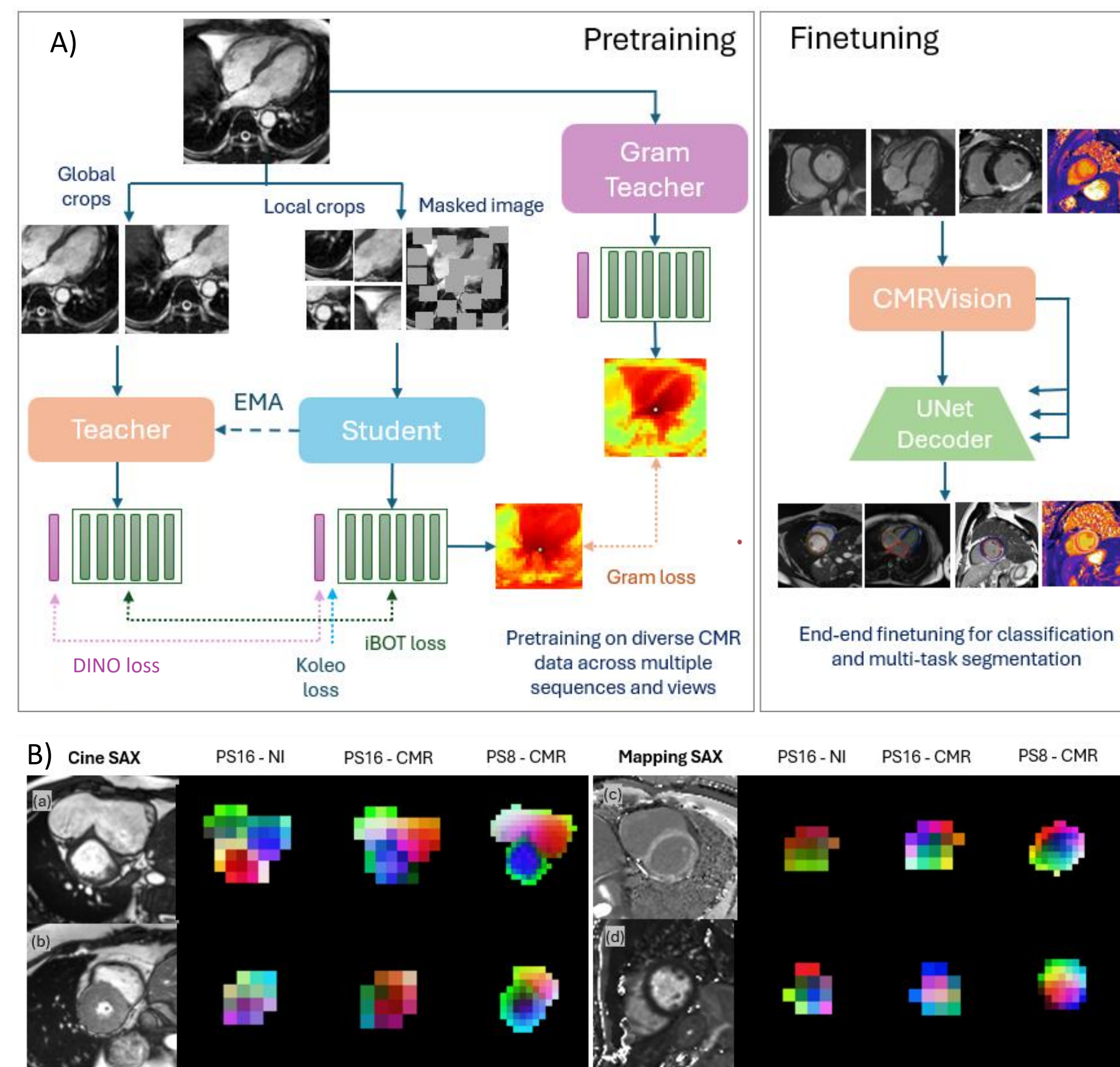


Figure:
A) Overview of the pretraining method

B) PCA visualizations of the feature space (masked by foreground mask) from pretrained encoders, pretrained on natural images (NI) with PS16 vs cardiac MRI (CMR) images with PS 16 and PS 8 - a) Cine SAX with dilated RV b) cine SAX with thickened myocardium, c) post-contrast T1 Mapping SAX d) T2 Mapping SAX. PS: Patch size

Results

Table 1. Segmentation results on the test set (Dice). (a) comparison of pretraining strategies; (b) comparison against general/CMR FMs and supervised baselines. Within each block, the best average Dice is in bold and the 2nd best is underlined. All methods are trained in the segment-all setting. † Method required additional class weighting for view-dependent structures for convergence.

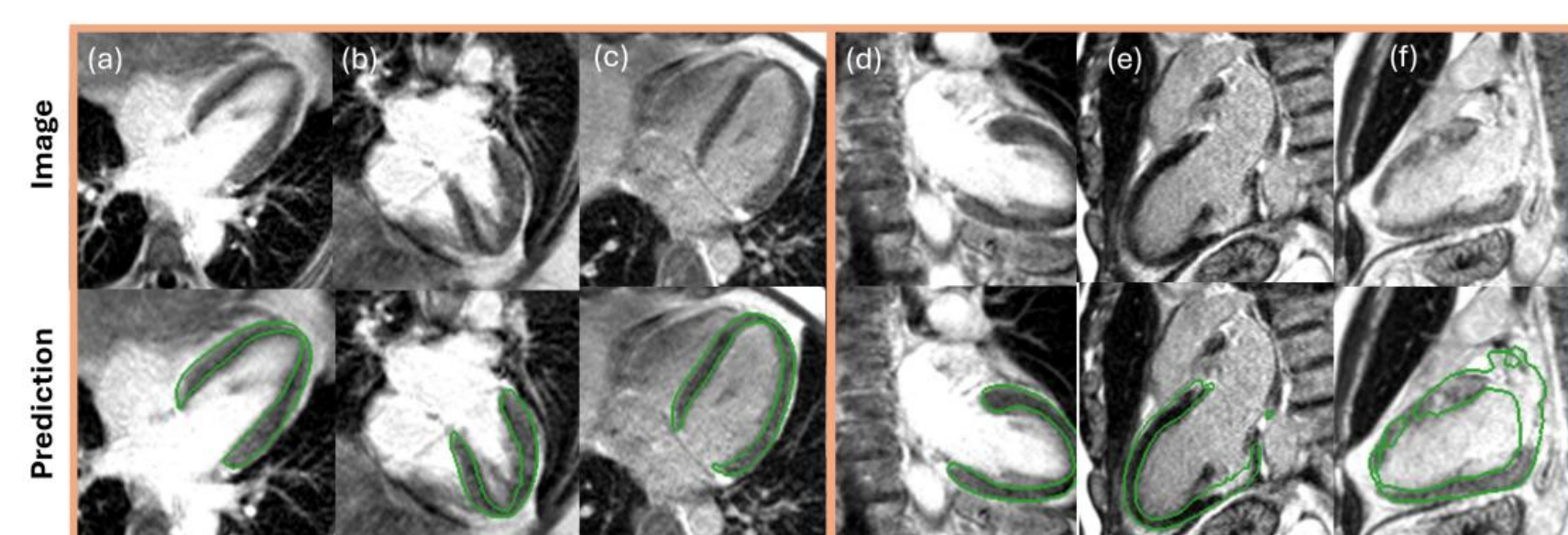
Method	Task Dice										
	Cine SAX			Cine LAX				LGE SAX		Mapping SAX	
	LV	Myo.	RV	LV	Myo.	LA	RA	LV	Myo.	LV	Myo.
<i>a) Comparison of pretraining strategies</i>											
DINOv3 (S/16, NI)	0.953	0.892	0.924	0.943	0.841	0.901	0.906	0.929	0.861	0.960	0.824
DINOv3/LoRA (S/16, NI)	0.955	0.898	0.925	0.949	0.851	0.910	0.916	0.927	0.863	0.963	0.840
DINOv2 (S/8, CMR)	0.961	0.906	0.926	0.954	0.860	0.916	0.925	0.930	0.877	<u>0.966</u>	<u>0.848</u>
DINOv3 (S/16, CMR)	<u>0.960</u>	0.904	0.930	<u>0.954</u>	<u>0.861</u>	<u>0.916</u>	<u>0.923</u>	<u>0.936</u>	<u>0.882</u>	0.965	0.846
CMRVision (S/8, CMR)	<u>0.960</u>	<u>0.905</u>	<u>0.929</u>	0.958	0.868	0.920	0.931	0.940	0.888	0.967	0.855
<i>b) Comparison against FMs and supervised baselines</i>											
SAM2	0.819	0.742	0.857	0.811	0.361	0.890	0.883	0.840	0.723	0.804	0.455
MedSAM2	0.885	0.813	0.895	0.773	0.358	0.866	0.901	0.897	0.742	0.880	0.713
Unetr	0.952	0.879	0.870	0.929	0.826	0.846	0.866	0.923	0.855	0.960	0.824
SwinUnetr	<u>0.953</u>	0.895	0.916	0.938	0.839	0.892	0.882	0.931	0.866	0.961	0.834
Jacob et al [13]	0.960	<u>0.902</u>	0.922	0.951	0.858	0.914	0.921	0.936	0.876	0.966	<u>0.846</u>
CineMA [5] [†]	0.960	0.905	<u>0.928</u>	<u>0.953</u>	<u>0.859</u>	<u>0.915</u>	<u>0.928</u>	0.933	0.879	0.963	0.836
CMRVision	0.960	0.905	0.929	0.958	0.868	0.920	0.931	0.940	0.888	0.967	0.855

For dense tasks such as segmentation, proposed model offers small but consistent benefits
CMR-pretrained models > NI-pretrained, patch size 8 > patch size 16,
domain-specific pretraining > LoRA adaptation

Table 2. Cine view classification results on the test set (accuracy). (a) comparison of pretraining strategies; (b) comparison with other methods. Within each block, the best average accuracy is in bold and the 2nd best is underlined.

Method	Avg. accuracy	Class					
		SAX	2CH	3CH	4CH	Aorta	Other
<i>a) Comparison of pretraining strategies</i>							
DINOv3 (ViTS/16, NI)	0.801	0.953	0.932	0.893	1.00	0.732	0.325
DINOv3/LoRA (ViTS/16, NI)	0.815	0.961	0.818	0.964	0.962	0.561	0.625
DINOv2 (ViTS/8, CMR)	0.908	0.930	1.00	0.964	1.00	0.829	0.725
DINOv3 (ViTS/16, CMR)	0.879	0.945	1.00	1.00	1.00	0.781	0.550
CMRVision (ViTS/8, CMR)	0.906	0.945	1.00	0.964	1.00	0.854	0.675
<i>b) Comparison with other methods</i>							
ViTS/8	0.707	0.813	0.705	0.929	1.00	0.195	0.600
SwinViT-Tiny	0.589	0.688	0.796	0.464	0.962	0.024	0.600
Jacob et al [13]	0.894	0.922	1.00	0.964	1.00	0.902	0.575
CineMA [5]	0.746	0.953	1.00	0.964	0.962	0.098	0.500
CMRVision (ours)	0.906	0.945	1.00	0.964	1.00	0.854	0.675

Average Dice score of 0.692, with scores of 0.612 and 0.794 on 2CH and 4CH views, respectively, indicating effective transfer to previously unseen anatomical viewpoints.



[1] Siméoni, O., Vo, H.V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., et al.: DINOv3. arXiv preprint arXiv:2508.10104 (2025)

[5] Fu, Y., Bai, W., Yi, W., Manisty, C., Bhuva, A.N., Treibel, T.A., Moon, J.C., Clarkson, M.J., Davies, R.H., Hu, Y.: A versatile foundation model for cine cardiac magnetic resonance image analysis tasks. arXiv preprint arXiv:2506.00679 (2025)

[13] Jacob, A.J., Borgohain, I., Chitiboi, T., Sharma, P., Comaniciu, D., Rueckert, D.: Towards a cardiovascular magnetic resonance foundation model for multi-task cardiac image analysis. Journal of Cardiovascular Magnetic Resonance 27(2), 101967 (2025)